Ion - Electron method

This method is used for ionic redox reactions.

Step 1

Using oxidation number concept, find out the reactants which undergo oxidation and reduction.

Step 2

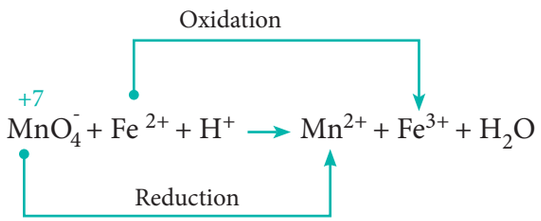
Write two separate half equations for oxidation and reduction reaction,

Let us consider the same example which we have already discussed in oxidation number method.

$KMnO_4 + FeSO_4 + H_2SO_4 \rightarrow$

$MnSO_4 + Fe_2(SO_4)_3 + K_2SO_4 + H_2O$

The ionic form of this reaction is,



The two half reactions are,

$Fe^{2+} \rightarrow Fe^{3+} + 1e^-$ ---------- (1)

and

$MnO_4^- + 5e^- \rightarrow Mn^{2+}$ ---------- (2)

Balance the atoms and charges on both sides of the half reactions.

Equation (1)$\Rightarrow$ No changes i.e.,

$Fe^{2+} \rightarrow Fe^{3+} + 1e^-$ ---------- (1)

Equation (2)$\Rightarrow$ 4'O' on the reactant side, therefore add $4H_2O$ on the product side, to balance 'H' - add, $8H^+$ in the reactant side

$MnO_4^- + 5e^- + 8H^+ \rightarrow Mn^{2+} + 4H_2O$---- (3)

Step 3

Equate both half reactions such that the number of electrons lost is equal to number of electrons gained.

Addition of two half reactions gives the balanced equation represented by equation (6).

(1) x 5 $\Rightarrow$ $5Fe^{2+} \rightarrow 5Fe^{3+} + 5e^-$ ---------- (4)

(3) x 1 $\Rightarrow$ $MnO_4^- + 5e^- + 8H^+ \rightarrow Mn^{2+} + 4H_2O$ ----------(5)

(4) + (5) $5Fe^{2+} + MnO_4^- + 8H^+ \rightarrow 5Fe^{3+} + Mn^{2+} + 4H_2O$----------(6)

Evaluate Yourself

8) Balance the following equation using oxidation number method

$As_2S_3 + HNO_3 + H_2O \rightarrow H_3AsO_4 + H_2SO_4 + NO$

11th Std Chemistry 001-037.indd 26 06-01-2021 15:02:55